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GRADO EN INTELIGENCIA Y ANALÍTICA DE NEGOCIOS

Inflationary inequality in Spain (2006-2025):
compositional mechanism, aggregation bias and
distributional effect of the VAT reduction on food

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Summary

This paper studies inflationary inequality among Spanish households according to their expenditure structure, using the Household Budget Survey and the Consumer Price Index at the level of Classification of Individual Consumption by Purpose detailed, from 2006 to 2025. An inflationary gap between expenditure quintiles is documented, explained by the composition of the consumption basket, and it is confirmed that aggregate price levels underestimate this heterogeneity (aggregation bias). The simulation of the VAT reduction on basic foodstuffs (2023-2024) shows a progressive but modest distributional effect compared to the magnitude of the inflationary shock.

Keywords

inflationary inequality, household inflation, aggregation bias, Household Budget Survey, fiscal policy



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1. Introduction

Since the beginning of 2021, inflation has shown an increasing and sustained trend on a global scale, reaching rates not seen in Spain in 35 years (Bank of Spain, 2022). The upturn placed the year-on-year rate of change of the Spanish CPI above 10% in some months of 2022, a level not seen since the 1980s. This episode revived a debate already present in the economic literature: whether the burden of inflation falls equally on all households, or whether, on the contrary, there are groups of the population systematically more exposed to its effects.

Between January 2006 and December 2025, the Spanish CPI accumulated a considerable variation, which hides, however, a notable disparity between the categories of expenditure that make up the consumption basket. As Table 1 shows, the cumulative variation by COICOP category ranged from 126.3% in Alcoholic beverages and tobacco, to a fall of 35.7% in Information and communications (the only category with a negative variation throughout the period), with the rest in an intermediate range ranging from 18.8% for Medicine, to 79.9% for Insurance and financial services, a category that, together with Personal Care, is broken down here by using the INE tool the ECOICOP classification see. 2 (base 2025), while the rest of this work, based on 2021, presents them unified under "Other goods and services".

Table 1. Cumulative variation of the CPI in Spain by COICOP (2006-2025)

Category	Percent Change (%)
Food and non-alcoholic beverages	72,9
Alcoholic beverages and tobacco	126,3
Clothing and footwear	29,1
Housing	65,3
Kitchenware	29,4
Medicine	18,8
Transportation	49,9
Information and communications	-35,7
Leisure and culture	34,9
Teaching	60,3
Hotels, cafes and restaurants	66,3
Insurance and financial services	79,9
Personal Care	45,4

Source: Prepared by the authors based on the Calculation of CPI Variations (INE), ECOICOP classification see. 2.

This heterogeneity between expenditure categories is precisely the mechanism that gives rise to the inflationary inequality that this paper studies. If households at different levels of expenditure allocate very different proportions of their budget to these categories, the same aggregate evolution of the CPI can translate into very different inflation experiences depending on the expenditure quintile in which each household is located. Basso, Dimakou and Pidkuyko (2022), in a study by the Bank of Spain, use the Household Budget Survey to construct a differentiated inflation rate according to the income, age and educational level of the household between 2006 and 2021, documenting that households in the lowest income quartile experienced systematically higher inflation than households in the highest quartile, with a gap that widens significantly during crisis episodes such as the one in 2021.

The Bank of Spain itself points out that, given that the prices of different goods and services do not evolve equally, and no household consumes exactly the same basket as "the representative household" of the aggregate CPI, each household experiences its own "personal" inflation, which depends on its particular consumption basket. This heterogeneity is directly relevant for the design of indexation and indirect taxation policies. In fact, the same study by Basso, Dimakou and Pidkuyko (2022) illustrates how fiscal policies in response to inflation can have very different distributional effects from each other: while certain indirect tax cuts benefit low-income households proportionally more, other support measures may be regressive.

However, the usual measures of inflation, such as the year-on-year rate of change of the headline CPI, are aggregate measures that do not reflect this heterogeneity between consumption baskets, since they are constructed from the expenditure pattern of the aforementioned "representative household". It is therefore necessary to construct an alternative measure that does reflect it, and that helps to design indexation policies in a more informed and equitable way. This heterogeneity, measured as the inflation gap between the lowest and highest spending quintile, is called "inflationary inequality" in the literature.

Although there is abundant international literature on inflationary inequality (Jaravel, 2021; Kaplan & Schulhofer-Wohl, 2017), studies focused on Spain are scarcer, and the work of Basso, Dimakou, and Pidkuyko (2022) itself is limited to a more aggregated level of product aggregation (2-digit COICOP) than that used in this work. This scarcity of Spanish literature is striking when compared to the already consolidated tradition in the United States: where studies such as those by Kaplan and Schulhofer-Wohl (2017) have been able to exploit scanner data at the barcode level. This Final Degree Project

contributes to this incipient literature by applying an already internationally validated methodology to a national case with a long time series (2006-2025) that covers several key Spanish economic cycles (the global financial crisis, the euro crisis and the post-pandemic energy crisis), and exploiting a finer level of product disaggregation (COICOP-4) than that used so far in the Spanish literature on the subject.

This level of detail is not an arbitrary choice: recent literature (Jaravel, 2021) documents an "aggregation bias", according to which working with aggregate price categories underestimates true inflationary heterogeneity between households, by averaging very different consumption patterns internally within the same category. This phenomenon is closely linked to the compositional mechanism that constitutes the central axis of this work: if inflationary inequality originates, as argued here, in the different composition of the consumption basket between quintiles, then the more aggregated the categories of expenditure used to measure that composition, the less capacity the analysis will have to distinguish different consumption patterns. Exploiting the deeper level of detail with the available data, i.e., the COICOP-4, therefore offers a more accurate estimate of real inflationary inequality.

Finally, this paper takes advantage of the fact that, within the study period itself, a real policy was implemented as a response to the inflationary crisis: the reduction of VAT on basic foodstuffs (RDL 20/2022). This allows us to go beyond the purely hypothetical simulation and build, with real data and an assumption of translation to prices calibrated from the existing literature, a counterfactual scenario that approximates to what extent and in what way a specific fiscal policy response could mitigate documented inflationary inequality, a particularly relevant question given that, as has been pointed out, the evolution of the price of basic necessities disproportionately affects lower-income households.

The research question that guides this work is the following: how much does inflationary inequality vary between Spanish households over time, where does this variation come from, and what policies could reduce it? To answer it, three specific objectives are proposed: (1) to measure inflationary inequality and its evolution between 2006 and 2025; (2) to explain the mechanism that causes it, and to verify to what extent the level of aggregation of the data used conditions its magnitude; and (3) to simulate the effect of a real fiscal policy on this inequality.

The rest of the work is structured as follows. Chapter 2 reviews the theoretical framework and the relevant literature. Chapter 3 describes the data and the methodology used. Chapter 4 presents the results, organized into four sections: the evolution of inflationary

inequality and its econometric models, the mechanism that explains it, the aggregation bias derived from the level of detail of the data, and the simulation of the VAT reduction on basic foodstuffs. Chapter 5 collects the conclusions, limitations and future lines of research of the work.

2. Theoretical framework

2.1. Evolution of inflationary inequality

The CPI (Consumer Price Index or Consumer Price Index) is an aggregate economic indicator that measures the monthly and annual variation in the prices of a basket of goods and services representative of household consumption. It is mainly used to calculate inflation and see how the cost of living changes. There is extensive literature on how inflation does not affect the entire population in a homogeneous way, although theoretically it is measured according to a consumption basket representative of the average household. This phenomenon is called "inflationary inequality", and various studies indicate that variables such as income, age or the level of education of the main breadwinner of the household play a relevant role in its magnitude and direction.

Along these lines, Basso, Dimakou and Pidkuyko (2022), in a study by the Bank of Spain, use the Household Budget Survey to construct a differentiated inflation rate according to income, age and educational level of the household between 2006 and 2021. These authors documented an average annual inflation of 1.7%. This value was 1.8% among households in the lowest income quartile, almost 0.2 percentage points higher than those with the highest income. Accumulated over the sixteen years, assuming that households remain in the same income bracket, this inflation gap rises to more than 2.75 p.p. Specifically, in 2021 inflation was particularly sensitive to income, with an average inflation of 2 p.p. higher among the quartile with the lowest expenditure compared to the quartile with the highest spending, which implies real inflation almost 30% higher among lower-income households.

On the other hand, Jaravel (2021) combines data from the U.S. *Consumer Expenditure Survey* with price data from the United States between 2004 and 2015. This author calculates an average annual inflation of 1.86% for the last income quintile, compared to 2.20% for the first. The average gap between the first and last quintiles was 0.346 percentage points. Focusing on the COVID-19 crisis, specifically during the May 2020 lockdown, households in the first quintile experienced an annualized inflation rate of 1.12%, compared to only 0.57% for higher-income households, i.e., a gap of 0.55 p.p.

2.2. Mechanism: Engel's Law and the Composition of the Basket

Both *works* mentioned attribute inflationary inequality to the combination of two facts: households of different incomes allocate different proportions of their expenditure to each category of consumption, and these categories do not experience the same inflation. Engel (1857) was the first to formalize the relationship between income and expenditure structure: "The prosperity of a people is determined by average consumption. The lower the percentage of all expenditure that must be allocated to food and physical sustenance, the richer the people will be, and vice versa."

Transferred to the household level, Engel's law implies that, although absolute expenditure on basic products increases with income, their proportion of total expenditure decreases as the household has higher income/expenditure. In short, each household has a different consumption basket, and this basket varies systematically with its income level. This is the motivation for the mechanism analysed: if the prices of the different categories of consumption evolve unevenly, this heterogeneity in the composition of expenditure translates into different inflation experiences depending on the level of household income.

For example, Basso, Dimakou and Pidkuyko (2022) document on the one hand that almost half of the average inflation in Spain between 2006 and 2021 was attributed to food and housing. Specifically, 21% of inflation came from food (food and non-alcoholic beverages), and 28% came from housing (including rent, imputed rent, electricity and gas); while transport only contributed 17%, and restaurants and hotels 11%.

On the other hand, they document an estimated coefficient of income over the proportion allocated to food of -0.056, and to housing of -0.036, while the estimated coefficient of income over the proportion allocated to transport is 0.029 and that of catering is 0.033, all significant. In short, as can be deduced with the help of Table 2, this results in a negative correlation between inflation and income, i.e., the higher the household income, the lower inflation they suffer, since lower-income households allocate a higher percentage of their budget to the products that contribute the most to inflation.

Table 2. Contribution to inflation and estimated income ratio to the proportion of expenditure, by relevant category (2006-2021)

Category	Contribution to inflation (%)	Income ratio to % of expenditure
Feeding	21	-0,056
Housing	28	-0,036
Transportation	17	0,029
Catering	11	0,033

Source: Authors' elaboration based on the results of Basso, Dimakou and Pidkuyko (2022).

2.3. Aggregation bias

Accurately measuring this heterogeneity according to income, however, requires more than just acknowledging that baskets differ between households: the level of detail with which price developments are measured within each consumption category also matters. The CPI, after all, was not designed to capture differences in exposure to inflation between individual households, but to summarize the aggregate evolution of prices in the economy; for the purpose of this paper, it is necessary to go further and work with the maximum level of product disaggregation available. Jaravel (2021) demonstrates why: the higher the level of product aggregation used to calculate the inflation experienced by each household, the greater the underestimation of real inflationary inequality between households. He calls this phenomenon "aggregation bias".

According to Jaravel (2021), older studies on inflationary inequality tend to find a modest effect on inflationary inequality, while more recent ones document substantial differences between income groups. This discrepancy is largely explained by the availability of data: recent studies have been able to work with a much higher level of output disaggregation, either through more detailed consumer surveys or, more recently, *scanner data*, which makes it possible to mitigate aggregation bias.

The U.S. Consumer Expenditure Survey used by Jaravel (2019) can be disaggregated to different levels according to COICOP, reaching a maximum detail total of 256 product categories. Using this level of disaggregation, the author compares the inflation gap between the first and fifth income quintiles, finding that the gap increases by more than 400%, from 0.07 to 0.35, going from 22 to 256 categories as seen in Table 3.

Table 3. Aggregation bias according to the level of detail of the COICOP classification

Aggregation level	Δ Inflation rate	
	Percentage points	% Explained
Detailed Categories N = 256	0,3464	100
Subcategories N = 22	0,0739	21,3
Main categories N = 11	0,0965	27,8

Source: Adapted from Jaravel (2019, p. 740).

However, even the 256 categories are still a product aggregation against the maximum possible level of detail: the price of each individual item, identified by its barcode (*scanner data*). An example of this type of data widely used in the literature on inflationary inequality is the Kilts-Nielsen Consumer Panel (KNCP), which tracks the shopping habits of approximately 50,000 households in the United States between 2004 and 2013, recording nearly 500 million transactions and around 1.4 million unique barcodes (UPCs).

According to Kaplan and Schulhofer-Wohl (2017), inflation measured with these data is substantially heterogeneous, with an interquartile range of between 6.2 and 9.0 p.p. per year, about five times higher than in studies with aggregated categories. Most of this variation comes from the different prices paid within the same type of product (for example, between brands or establishments), and not from differences in which product categories each household buys. This is not a different mechanism than the composition of the basket, but the same source of heterogeneity taken to its finest level of detail: while at the level of COICOP categories the differential composition is observed between types of product (milk versus cheese, for example), at the barcode level it is observed within the same type of product (which brand of milk each household buys), something that only the barcode detail allows us to capture. The underlying mechanism, however, follows the usual pattern in the literature: a difference of almost 1 p.p. per year between households in the first and last quintiles, which could even be underestimated anyway, since these Nielsen data do not cover categories such as education, health or gasoline, which are relevant to inflationary inequality (Hobijn and Lagakos, 2003, as cited in Kaplan and Schulhofer-Wohl, 2017).

The numerical comparisons established in this paper with Basso, Dimakou and Pidkuyko (2022), Jaravel (2019, 2021) and Kaplan and Schulhofer-Wohl (2017) should be interpreted with due methodological caution. These studies differ from the present work

in at least four aspects: the stratification variable used to construct the comparison groups (gross income in Basso et al. and López Laborda et al., versus the equivalent expenditure used here); the number of groups (quartiles in Basso et al. and Almunia et al., versus quintiles in this work); the time period covered (2006-2021 in Basso et al., 2004-2015 in Jaravel, 2004-2013 in Kaplan and Schulhofer-Wohl, versus 2006-2025 here); and the source of microdata and the economic context (the United States in Jaravel and in Kaplan and Schulhofer-Wohl, Spain in the rest). For these reasons, the magnitudes that are compared throughout this work are presented as contrasting references that make it possible to assess whether the orders of magnitude obtained are reasonable in the light of the existing literature, and not as directly equivalent figures obtained under identical methodological conditions.

2.4. Fiscal policies and their effect on inflationary inequality

Ignoring this inequality (its magnitude, its mechanism, and the biases that underestimate it) has direct consequences for the design of public policy: without this information, the State runs the risk of allocating resources inefficiently, without adequately protecting the most vulnerable households. This motivates assessing whether fiscal responses to the inflationary crisis have taken this heterogeneity into account.

Basso, Dimakou and Pidkuyko (2022) assess the distributional effect of a Spanish real fiscal policy: the reduction in VAT and excise duties on electricity and gas approved in the summer of 2021 (RDL 12/2021 and RDL 17/2021). They find that, without the tax cut, the contribution of electricity to inflation in June 2022 for households in the poorest quartile would have been 3.5 p.p., compared to the 2.2 p.p. observed with the current reduction (on a total inflation of this group of approximately 8.5 p.p.); for households in the richest quartile, this contribution would have been 1.9 p.p. compared to 1.2 p.p. observed. The gap between the two quartiles in the specific contribution of electricity was therefore reduced from 1.6 p.p. without the reduction to 1 p.p. with it, showing that the measure benefited lower-income households proportionally more.

On the other hand, the policy on which this paper focuses, the extraordinary VAT reduction on basic foodstuffs (RDL 20/2022), has been causally evaluated by Almunia, Martínez and Martínez (2023). Using daily online price data from three large Spanish supermarket chains (with a combined market share of 40%), the authors compare the price evolution of the products affected by the rebate (staple foods, whose VAT fell from 4% to 0%, and oil and pasta, from 10% to 5%) against a control group of similar products not affected, using a difference-in-differences model. To measure the distributional effect,

they combine the estimates of translation to prices with microdata from the EPF (2021) broken down by expenditure decile, an approach similar to that of this work.

They find that the measure was regressive in absolute terms: annual savings ranged from €35 for lowest-spending households to €85 for highest-spending households, and 50% of the revenue not received by the State benefited the four deciles with the highest economic capacity; but progressive in relative terms: savings represented 0.3% of the total annual expenditure of households in the bottom decile, compared to only 0.13% in the upper decile. This pattern is a direct reflection of Engel's law: higher-income households spend more in euros on food (hence their higher absolute savings), but that amount represents a smaller proportion of their total budget (hence their lower relative savings). The authors conclude that the impact of the measure was, in any case, modest compared to the magnitude of the inflationary shock it was intended to alleviate, and that a targeted income transfer would have been more efficient from a distributional and budgetary point of view.

The same policy has also been evaluated by López Laborda, Marín González and Onrubia (2023), with a different methodological approach: instead of observed market prices, they statistically merge the microdata from the EPF and the Living Conditions Survey to simultaneously have income and expenditure information for each household. Under the assumption of full translation (of 100%) to prices, they estimate that the average savings per household ranged from €22.4 for the lowest gross income quintile to €60.7 for the top percentile, during the initial six months of the measure's validity, with a total collection cost of €700 million, of which half benefited the 40% of highest-income households, a figure that illustrates, once again, the regressive nature in absolute terms shared by this type of universal tax reduction.

3. Methodology

3.1. Data sources: the Household Budget Survey and the CPI

The two sources of this work are the Household Budget Survey (EPF) and the annual CPI, both from the National Institute of Statistics, for the period 2006-2025.

The EPF is an annual survey that collects consumer expenditure data from nearly 24,000 households, with three main purposes: to update the weights of products in the CPI, to estimate the private consumption of the economy for the National Accounts, and to serve as a basis for the social analysis that guides the design of social and fiscal policies.

This survey has been carried out with different classifications of goods and services depending on the period. Between 2006 and 2015, the COICOP/EPF 2006, a national adaptation of the United Nations international classification, was used; between 2016 and 2023, the ECOICOP ver.1, the European harmonized version. Both classifications coincide up to the level of detail used in this work (4 digits), so it was not necessary to apply any additional correlation table for this first section of the series.

From 2024, however, the new United Nations international consumer classification, called COICOP/EPF 2018, will be incorporated. The main difference with the classification that has been used since 2016 (ECOICOP ver.1) is that it extends the previous 12 categories of expenditure to 13, so that expenditure on insurance and financial services forms a separate group.

To standardise this coding with the rest of the series, the official table of correspondences between COICOP 2018 and ECOICOP ver.1 published by the INE was used. It should be noted that this correspondence is not uniformly one-to-one: in some cases, the expenditure of a source code is distributed, according to a coefficient published by the INE, between two different target categories, and vice versa (for example, "Health-related insurance" comes 90% from "Health insurance" and 10% from "Accident insurance").

With household expenditure already classified homogeneously, the second source of this work is the Consumer Price Index, which, according to the CPI methodology published by the INE (National Institute of Statistics, n.d.-b), is calculated by visiting approximately 33,000 establishments (the most frequented in all of Spain) and collecting the prices of the best-selling products in each one, until a total of almost 220,000 prices are observed. With these data, a "basket" of goods representative of household consumption is constructed, weighted according to the expenditure that they allocate to each one (using the EPF to calculate the weights).

This paper uses the 2021 base, whose basket consists of 955 articles (462 from traditional collection and the rest from scanner data). The change of base is a process that the INE carries out every five years, in order to renew the CPI methodology; technically, the most recent base is that of 2025, brought forward by the integration of ECOICOP ver.2. However, this classification is similar to that of the COICOP/EPF 2018, with 13 categories instead of 12, while the 2021 base uses ECOICOP ver.1, in line with the classification adopted in the EPF for most of the study period. For this reason, consistency with the EPF is prioritized over the actuality of the CPI base.

Within this classification, the INE assigns each subclass a weight proportional to the percentage of the budget that households allocate to its purchase, calculated from the EPF. Formally, according to the National Institute of Statistics (2026-a), the weight of the item w_i is defined as:

$$w_i = \frac{\text{expenditure on the subclasses represented by the item } i}{\text{total expenditure}}$$

3.2. Construction of inflation by household

With these sources, the different data tables that make up the sample used are constructed. On the one hand, from the EPF, tables of individuals, detailed expenses, aggregate expenses and households are created. The table of individuals includes personal information of the members of each household: relationship with the main breadwinner, age and sex. The expenditure table includes the annual expenditure of each household (once its weighting has been undone) and its COICOP code, separating it into detailed (disaggregated up to the COICOP-4 code) and aggregate (at the COICOP-1 level). Finally, the household table includes the elevation factor of each household and the Autonomous Community to which it belongs, together with the GID-1 code of the GADM regions, a global unique identifier for the first-level administrative subdivisions. Most of the sociodemographic variables collected are not used directly in this work, but they are preserved to facilitate possible future extensions that explore inflationary inequality according to these characteristics, without the need to repeat the process of cleaning the EPF microdata.

The complete code used for microdata cleaning, household inflation construction, and policy simulation is publicly available (Del Olmo Tirado, 2026).

As a check for the data cleansing, the average annual expenditure per household calculated from the microdata was compared with the official figures published by the INE. As can be seen in Table A1, the coincidence is exact for the period 2016-2025, and practically exact (average difference of -0.13%) for 2006-2015. The INE itself documents that the period 2006-2015 has been the subject of successive methodological revisions, both of the classification of goods and services, and of the reference population used in the elevation factors, updated from the 2001 Census to the 2011 Census in March 2013 (National Institute of Statistics, n.d.-a), which may explain the small discrepancy observed between the microdata used in this work and the aggregate reference figures. This discrepancy, of insignificant magnitude for the purposes of this work, validates the correct construction of the expenditure variable from the microdata.

In addition, the detailed and aggregate inflation tables are constructed. For the aggregate version, the monthly CPI of each category is added and divided by 12; then, the inflation of a certain category and year $I_{c,t}$ is calculated according to the following formula:

$$I_{c,t} = \sum \frac{IPC_{c,t}}{IPC_{c,t-1}} - 1$$

A limitation of the inflation table is that the EPF distinguishes more categories of expenditure than those published by the CPI with its own price at the COICOP-4 level: of the 430 COICOP-4 combinations with expenditure observed in the EPF throughout the series, 224 (52%) lack an equivalent category in the CPI, a gap that remains stable in all years of the period (between 129 and 154 codes without correspondence per year), and which is therefore not due to incomplete coverage in specific years, but to the fact that the INE groups, for the purposes of price collection, several COICOP-4 subclasses under the same representative article, and to the fact that certain items are not measured by visits to establishments, the usual method of collecting the CPI, but by specific administrative procedures that do not generate an item with its own price at that level of detail. Among the latter, imputed rent stands out in particular (04.2): its COICOP-4 codes lack, without exception, an item with its own price in the CPI, a result consistent with what Basso, Dimakou and Pidkuyko (2022) point out, who warn that the INE does not incorporate the variation in the imputed cost of home ownership as part of the expenditure whose inflation is measured by the CPI, precisely because there is not even an observable price on which to build a price index for this item.

To resolve this discrepancy, the data at the lowest level and at level 4 were downloaded separately, applying a cascade combination method: for each category of the EPF, its equivalent in the CPI at level 4 is first sought; if it does not exist, the next higher level 3 is used, etc. This methodology implies assuming that inflation is identical between them, which introduces a loss of precision, which is anticipated to affect the results of the aggregation bias. Even so, it makes it possible to keep the information of the products and years, which do have level 4 available.

With these tables, we proceed to construct inflation per household, calculated as the weighted sum of the inflation of each expenditure category, using as a weighting the proportion of household expenditure allocated to each one. Formally, household inflation I_h is defined as:

$$I_h = \sum (Prop_{h,c} \times I_c)$$

where is the proportion of household expenditure $Prop_{h,c}$ allocated to each category c and I_c is the inflation of that category. This formula assumes that each household's basket remains fixed within each year, without capturing the possible substitution of products in response to relative price changes, a limitation that is taken up in the conclusions.

3.3. Equivalence scale and quintiles of expenditure

Once inflation per household has been calculated, households are grouped into quintiles according to their equivalent per capita expenditure. It is worth mentioning that part of the literature on inflationary inequality uses income, rather than expenditure, to construct quantiles; in this paper, on the other hand, expenditure has been chosen for two reasons. On the one hand, it is the variable used by the INE itself to construct the quintiles in its EPF reports. On the other hand, income does not always accurately reflect the standard of living or the real consumption pattern of a household, since two households with the same income can allocate very different proportions of it to savings versus expenditure, a proportion that, moreover, tends to vary systematically with the level of income, with the lower the share of expenditure being greater due to the inability to save. Expenditure, by directly capturing what the household actually consumes, is a more direct basis for classifying households according to their exposure to inflation.

In addition, instead of simply dividing expenditure by the number of household members, the OECD equivalence scale commonly used by the INE was used, which makes it easier to compare expenditure between households of different sizes and compositions.

Specifically, the following coefficients apply: 1 for the main breadwinner, 0.7 for the rest of the adults, and 0.5 for children under 14 years of age. The equivalent weight of a household P_h , with A adults (including the main breadwinner) and M children under 14, is calculated as:

$$P_h = 1 + (A - 1) \times 0.7 + M \times 0.5$$

The per capita equivalent household expenditure $G'_{percap,h}$ is then obtained by dividing the total household expenditure by this equivalent weight:

$$G'_{percap,h} = \frac{G_h}{P_h}$$

Once the equivalent per capita expenditure of each household has been calculated, households are ordered from lowest to highest according to this value. Equivalent

expenditure is used, and not the total expenditure of the unadjusted household to avoid the resulting order being skewed by economies of scale of consumption: since part of a household's expenditure is shared among its members (housing, supplies, certain appliances), a household does not need to spend twice as much for its four members to live as well as a single one, and sorting by total expenditure without adjusting would unfairly penalize large households over small ones.

However, ordering is not enough to decide where the cut is between one quintile and the next: it is necessary to accumulate, household by household, how much population is below each cut-off point, until reaching 20%, 40%, etc. To do this, the real population of each household (the product of its size times its elevation factor), and not its equivalent population, is deliberately used. This choice responds to the fact that equivalent households and real households do not correspond proportionally: households with lower expenditure tend to be, on average, more numerous, and since the equivalence weight of each additional member is less than 1 (0.7 or 0.5, as the case may be), their equivalent population is systematically lower than their real population. If equivalent population were accumulated instead of real population to determine the cuts, more households in the first quintile would be needed to complete the same equivalent population threshold, precisely because these households have more members on average, which implies that they weigh less in equivalent terms than in real terms; And since each of these additional households in turn contributes several real people, the cut-off point of the first quintile would be reached by dragging more real people than should actually be below it.

With these expenditure data by household and quintile, the inflation tables are finally combined to obtain the *complete dataframe* used in the rest of the work, whose main variables are: the household identifier, the quintile to which it belongs, the elevation factor, its inflation calculated with the detailed COICOP, and its inflation calculated with the aggregate COICOP.

An important feature of this approach is that inflation is calculated only at the level of the COICOP category: all households consuming the same category therefore receive the same inflation rate in that category, regardless of their expenditure quintile. This does not invalidate the inequality analysis, on the contrary, it allows precisely the underlying explanatory mechanism to work: since inflation in each category is common to all households, any difference in total inflation experienced by different quintiles must necessarily originate in their different composition of the consumption basket.

Based on the inflation of each household, the average inflation $\bar{I}_{q,t}$ of a quintile q in the year t is defined as the average inflation of its households, weighted by the elevation $FACTOR_h$ factor of each household h :

$$\bar{I}_{q,t} = \frac{\sum_{h \in q} I_{h,t} \times FACTOR_h}{\sum_{h \in q} FACTOR_h}$$

Where $I_{h,t}$ is household inflation and the sum extends to all households in the quintile q in the year t . The inflation gap between the quintile of lowest and highest expenditure in a year B_t , the main metric of this work, is defined as the difference between both average inflations:

$$B_t = \bar{I}_{q=1,t} - \bar{I}_{q=5,t}$$

A positive gap indicates that the lowest spending quintile suffered higher inflation that year; a negative gap, the opposite. Finally, the cumulative gap B'_T over a period of years T is defined, following the same accumulation logic used by Basso, Dimakou and Pidkuyko (2022), as the simple sum of the annual gaps for that period:

$$B'_T = \sum_{t=1}^T B_t$$

3.4. Econometric model

To test whether the differences in inflation by quintile are statistically significant, three regression models with fixed effects of the year are estimated, weighted by the elevation factor of each household, with standard errors robust to heteroskedasticity.

The first of the three models does not include any quintile variables, and serves as a reference to isolate how much variance in inflation per household explains, on its own, the macroeconomic cycle common to all quintiles each year:

$$I_{h,t} = \alpha_t + \varepsilon_{h,t}$$

The second estimates the average effect of belonging to each quintile throughout the period:

$$I_{h,t} = \alpha_t + \sum_{q=2}^5 \beta_q \times 1[QUINTIL_h = q] + \varepsilon_{h,t}$$

taking Quintile 1 as a reference category. The last model allows this effect to vary freely each year, with the quintile interacting with the year:

$$I_{h,t} = \alpha_t + \sum_{q=2}^5 \sum_t \beta_{q,t} \times \mathbb{1}[QUINTIL_h = q] \times \mathbb{1}[anio_t = t] + \varepsilon_{h,t}$$

The first model quantifies the part of the variance attributable only to the common macroeconomic cycle; the second, whether there is a net effect of the quintile accumulated over the entire period; and the third, whether this effect hides a year-to-year variability.

To assess the explanatory capacity of each model, two different adjustment measures are used. The *Adj. R²* measures the proportion of the total inflation variance of each household explained by the complete model, including the fixed year effect. The *Within R²*, on the other hand, measures the proportion of the variance that remains after discounting the fixed effect of the year, that is, the variation that occurs within each year between households, thus isolating the explanatory capacity of the variable of interest (the quintile, alone or interacted with the year), regardless of the year effect. Given the high sample size used (more than 420,000 observations), the statistical significance of a coefficient must be interpreted together with these two adjustment measures, and not in isolation: with samples of this size, even substantively small differences between quintiles can be significant, so a low p-value does not imply, by itself, economic relevance.

3.5. Mechanism behind inflationary inequality

To identify the mechanism behind inflationary inequality, two magnitudes are constructed from the maximum level of detail available (COICOP-4): the weight of each category in the consumption basket of each household, and the contribution of each category to the total inflation of that household.

The weight of each category is calculated, for each household, as the proportion of its total expenditure allocated to each combination of COICOP-4, the same variable $Prop_{h,c}$ used in the rest of this work to construct I_h , household inflation. Aggregated by quintile, this weight measures what proportion of the joint expenditure of all households in the quintile is allocated to each category, first adding the weighted expenditure of each household and then dividing by the total weighted expenditure of the quintile:

$$\overline{Prop}_{q,c} = \frac{\sum_{h \in q} G_{h,c} \times FACTOR_h}{\sum_{h \in q} \sum G_{h,c} \times FACTOR_h}$$

where q denotes the quintile, and c a specific COICOP-4 category.

The contribution to inflation is also calculated, multiplying the weight of each category by the inflation of that category in each year, and adding it back at the quintile level using the same weighted sum procedure:

$$\overline{Contr}_{q,c} = \frac{\sum_{h \in q} (Prop_{h,c} \times I_c) \times FACTOR_h}{\sum_{h \in q} FACTOR_h}$$

Both magnitudes are calculated entirely at the COICOP-4 level, and aggregated at COICOP-1 level, summing all the level 4 subcategories within each of the 12 divisions for visualization and interpretation purposes only, since representing all the level 4 categories in a single graph would be unreadable. This final aggregation does not imply any loss of precision with respect to the original calculation, since the expenditure and contribution of each household to each level 1 category preserves exactly the sum of its level 4 components.

3.6. Aggregation bias: comparison between levels of detail

To empirically contrast Jaravel's (2021) prediction that working with poorly disaggregated product categories underestimates real inflationary inequality, the inflation of each household was recalculated using the CPI at the COICOP-1 level (the usual level of aggregation in official inflation indicators), instead of the COICOP-4 detail used in the rest of this work. Both series are constructed from the same consumption basket of each household and the same aggregation process by quintile described in the general methodology, differing only in the level of disaggregation of the CPI used to calculate inflation for each category.

Based on these two series, the aggregate level is compared with the detailed level in two metrics: the Q1-Q5 gap, which is the main metric of this work, and the dispersion index among all households (the weighted standard deviation), which allows us to contrast whether the aggregation bias affects only the extreme quintiles or the distribution as a whole.

The weighted standard deviation was chosen instead of the coefficient of variation (CV) as a measure of dispersion since the latter is artificially triggered in years with average

inflation close to zero, a purely arithmetic problem of dividing by a denominator close to zero, unrelated to real inequality. The weighted variance is calculated, for each year, as

$$Var = \frac{\sum FACTOR_h \times (I_h - \bar{I})^2}{\sum FACTOR_h - 1},$$

where $\bar{I}$ is the weighted mean of inflation for all households that year and the weights are, again, the elevation factor of each household; this expression applies a correction analogous to Bessel's, substituting the sample size for the sum of the elevation factors, to obtain an approximately unbiased estimator of population variance. The weighted standard deviation is obtained, finally, as the square root of this variance.

For each year t , aggregation bias is quantified as the relative increase ΔB_t , in absolute value, of the detailed metric with respect to the aggregate metric:

$$\Delta B_t = \frac{|M_{det,t}| - |M_{agr,t}|}{|M_{agr,t}|}$$

where M represents, as the case may be, the Q1-Q5 gap or the weighted standard deviation, and the sub-indices *det* and *agr* distinguish the COICOP-4 (detailed) level from the COICOP-1 (aggregate). The absolute value of both magnitudes is taken before calculating the difference because what is relevant for the aggregation bias is the magnitude of the inequality detected at each level, regardless of its sign (whether Q1 or Q5 is more affected). The bias is summarized, throughout the series, with two figures: the proportion of years in which the detailed level exceeds the aggregate in magnitude, and the average of that percentage increase over the twenty years of the series. This average, however, must be interpreted with caution: in years when the aggregate gap is very close to zero, dividing by such a small number artificially triggers the percentage increase, so that a few specific years can considerably distort the overall figure, even if the real difference between the two levels is small that year.

3.7. Policy simulation: counterfactual without VAT reduction on basic foodstuffs

To assess the distributional effect of the VAT reduction on basic foodstuffs (RDL 20/2022), a counterfactual scenario was constructed that reconstructs the inflation per household that would have been observed in the absence of this policy. The logic of the exercise is as follows: given that the real inflation of 2023 and 2024 already incorporates the effect of the reduction (the measure was in force during both years), the inflation of the affected products was recalculated by adding back the price change that the

reduction prevented, thus obtaining the inflation that these products would have had if the measure had not been approved.

The affected products were identified at the COICOP-4 code level, comparing each subclass of the classification with the official list of products included in RDL 20/2022: staple foods (cereals, bread, milk, cheese, eggs, fresh fruit and vegetables, legumes and tubers), whose VAT rate went from 4% to 0%, and oil and pasta, whose rate went from 10% to 5%. The theoretical price change ΔP under a full pass-through assumption was calculated using the standard tax formula:

$$\Delta P = \frac{1 + \tau_1}{1 + \tau_0} - 1$$

where τ_0 and τ_1 are the pre- and post-reform tax rates, respectively. Given that the pass-through of a tax reduction to final prices is not necessarily complete, this theoretical change was weighted by the empirical pass-through rate of 90% estimated by Almunia, Martínez and Martínez (2023) for the Spanish case, based on the daily price data of large stores described.

It also incorporates the evolution of the tax rate throughout the period of validity of the measure, which did not remain constant: the total exemption (0% and 5%, respectively) was in force until 30 September 2024, after which a gradual de-escalation (2% and 7.5%) began until 31 December of the same year, the date on which the measure ended. As expenditure data with monthly granularity were not available, the theoretical price change for 2024 was calculated as a weighted average by the number of months in each bracket (nine months at the full rate, three at the de-escalation), while 2023 was calibrated with the full rate during the twelve months of the year.

Within this calendar, olive oil also received a specific tax treatment that was different from that of other oils and pasta: while seed oils and pasta maintained the reduced rate of 5% until 30 September 2024 (later climbing to 7.5% until 31 December), olive oil went from 5% to 0% between 1 July and 30 September 2024, and from there to 2% until 31 December of the same year (Royal Decree-Law 4/2024). This work therefore models three calendars of different types: basic foodstuffs, olive oil, and seed oils and pasta, each weighted by the number of months of validity of each tranche within 2024.

It should be noted that olive oil did experience, since January 1, 2025, a permanent reduction in its general rate (from 10% to 4%) approved by the same Royal Decree-Law 4/2024; it is, however, a non-temporary modification of the tax rate, unrelated to the object of this work, so it is not incorporated into the counterfactual scenario.

Regarding the rate of translation to prices used (90%), it should be stressed that this is a calibration assumption, not an estimate specific to this work: it comes from the causal assessment of Almunia, Martínez and Martínez (2023), who obtain it from daily prices of three supermarket chains during the first half of the measure's validity. This paper assumes that this rate remains stable for the set of products affected and for the entire period 2023-2024, a reasonable simplification assumption given the lack of an equivalent estimate for the rest of the period or for other types of establishment, but which should be understood as such, and not as a magnitude independently verified for this work.

It is important to emphasize that the formula used does not consist of arithmetically inverting the theoretical fall in prices on the observed inflation, but of reconstructing the counterfactual price that would have been observed if the rebate had not existed: to real inflation (which already incorporates the effect of the rebate) is added back the change in the gross price that the rebate prevented, weighted by the pass-through rate. In this way, the resulting counterfactual inflation represents the evolution of prices that could be expected under the tax rates in force before the reform, and not a simple algebraic transformation of the theoretical fall in prices.

Finally, given that the VAT reduction is not applicable in the territories with their own indirect taxation, i.e. the Canary Islands General Indirect Tax (IGIC) in the Canary Islands and the Tax on Production, Services and Imports (IPSI) in Ceuta and Melilla; households resident in these territories were excluded from the counterfactual scenario, maintaining the real inflation observed for them.

The resulting counterfactual scenario was applied to the consumption basket of each household, following the same quintile aggregation methodology used in the rest of this work, and compared with the real Q1-Q5 gap.

In addition, in order to express this effect in monetary terms, the additional cost in euros was first calculated at the level of each individual household, multiplying the difference in inflation between its real scenario and its counterfactual scenario by its own monthly expenditure, and then averaging this result, weighted by the elevation factor, within each quintile. This figure is contrasted, in chapter 4.5, with the estimates of savings per household by Almunia, Martínez and Martínez (2023), López Laborda, Marín González and Onrubia (2023) in order to assess the distributive capacity of the measure.

4. Results

4.1. Evolution of inflationary inequality

The average gap for the entire period (2006-2025) is 0.112 percentage points (SD = 0.691 p.p.), and its cumulative gap over the 20 years (adding the annual difference of each one) amounts to 2.24 p.p. This last figure, however, is a small magnitude compared to the variability of the series: the sign of the gap alternates frequently throughout the twenty years analysed, positive in most of them, but negative in 2013, 2014, 2015, 2016, 2019, 2020, 2023 and 2024, so that the average for the period is not particularly revealing. This alternation of sign is consistent with the absence of a sustained trend over the period, a pattern that is examined in greater detail in section 4.2, and suggests that inflationary inequality manifests itself as a succession of specific episodes, rather than as a constant regressive or progressive bias over time, as can be seen in Figure 1.

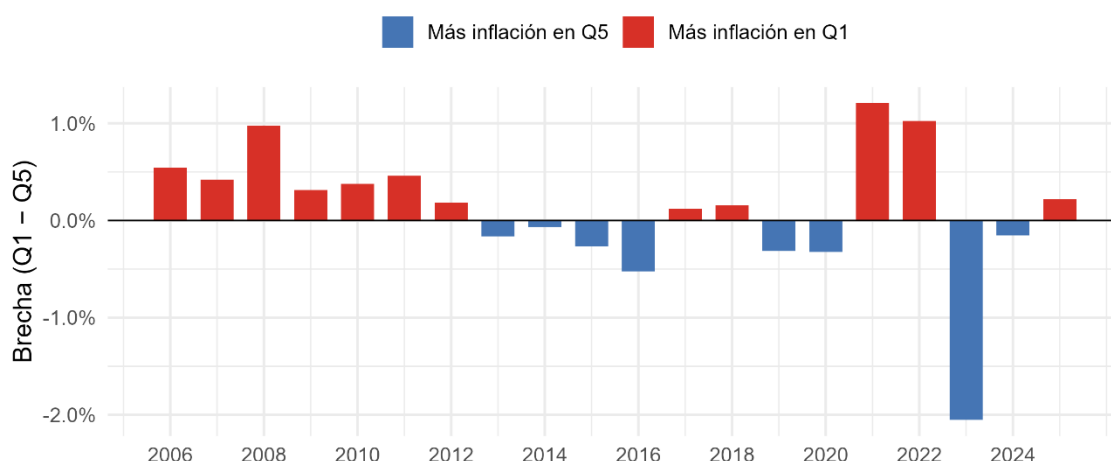


Figure 1. Inflation gap between the first and last quintile, in annual inflation

Restricting the analysis to the period 2006-2021 to compare it with Basso, Dimakou and Pidkuyko (2022), the average annual gap is 0.194 p.p., very similar to the 0.2 p.p. that these authors report for the same period. Accumulated over the 16 years, following the same accumulation logic as the authors, the gap amounts to 3.10 p.p., somewhat higher than the 2.75 p.p. reported by the authors for the extreme income quartiles. The differences could be attributable, among other methodological differences, to the use of quintiles in this work, compared to quartiles in the reference study, given that the quintiles capture a tail of the most extreme distribution and therefore a somewhat larger gap.

At the level of individual episodes, this paper identifies 2023 as the year with the highest sensitivity of inflation to spending, with a gap of -2.02 p.p., the value with the highest absolute magnitude in the entire series, followed by 2021 (1.20 p.p.) and 2022 (1.08

p.p.). Unlike 2021-2022, in 2023 it was the highest-spending households that suffered proportionally more inflation than those with the lowest spending. Basso, Dimakou and Pidkuyko (2022) report a gap of 2 p.p. for a displaced year-on-year window (April 2021 to March 2022) that does not coincide exactly with the calendar year 2021 analysed here (5.70% for Q1 and 4.50% for Q5, a gap of 1.20 p.p.), although the magnitude is, in this case, very close to theirs; Both studies coincide in pointing out 2021-2022 as an episode of high inflationary heterogeneity by income level, coinciding with the post-pandemic energy crisis, although in this work this episode is exceeded in magnitude by 2023, a year that the authors did not study.

Finally, restricting the analysis to the 2006-2015 subperiod, the closest to the 2004-2015 window used by Jaravel (2019) for the United States, we obtain an average inflation of 2.40% for the lowest expenditure quintile and 2.12% for the highest, levels close to those reported by Jaravel for its lower and upper quintile (2.20% and 1.86%, respectively). The resulting gap (0.277 p.p.) is, in this case, slightly lower than its own (0.346 p.p.), a magnitude of the same order despite the absence of the years 2004-2005 in the Spanish series.

Complementing the gap between the extreme quintiles, the weighted standard deviation of inflation among all households within each year was also calculated, a metric that, unlike the Q1-Q5 gap, is not limited to comparing the two most distant income groups, but summarizes the joint dispersion of the entire distribution of households.

The series, as can be seen in Figure 2, shows a pattern consistent with that observed in the Q1-Q5 gap: the lowest dispersion values are concentrated in years of moderate or near-zero headline inflation (2014, 2018 and 2019, with standard deviations of around 0.5 percentage points), while the steepest peaks coincide with the main episodes of the inflationary gap of the period: 2021 (2.10 p.p.), and more markedly 2022 (2.75 p.p.) and 2023 (3.23 p.p.).

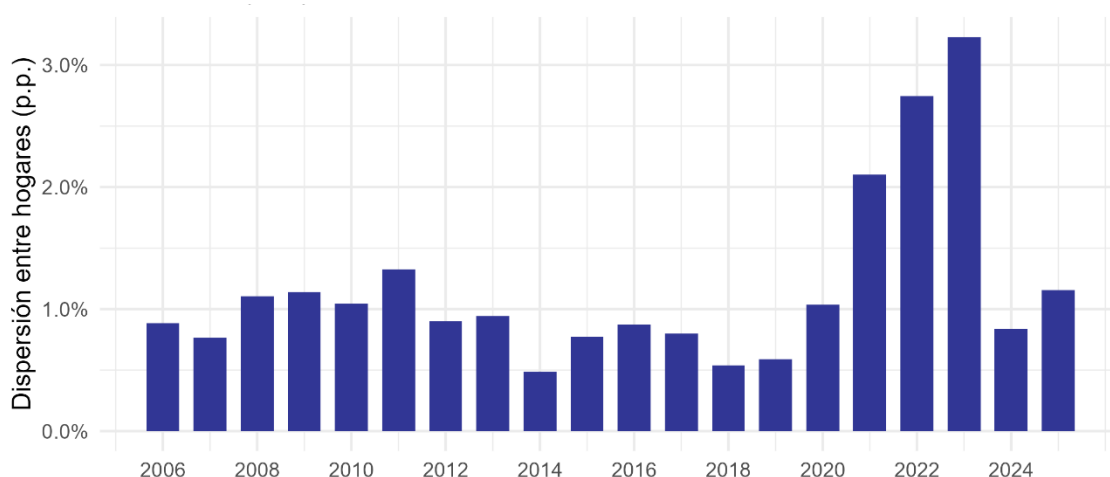


Figure 2. Weighted dispersion of inflation among all households, 2006-2025

The fact that 2023 registers the highest dispersion of the entire series is consistent with the fact that it is also the year with the highest magnitude of the Q1-Q5 gap: higher-spending households suffered significantly higher inflation that year than lower-spending households: both metrics reach their maximum values in the same year.

4.2. Econometric analysis of differences by quintile

Applying the aggregate model described in the methodology, a small net effect is documented: over the twenty years of the period 2006-2025, quintile 2 has an average inflation slightly higher than that of quintile 1 (+0.0234 p.p.,), quintile 3 does not differ significantly from quintile 1 (+0.0023 p.p.,), and both quintiles 4 and 5 have significantly lower average inflation (-0.0353 p.p. and -0.0986 p.p., respectively, in both cases with). The $p = 0,027$ $p = 0,824$ $p < 0,001$ Within R^2 (0.11%) is, however, very small, indicating that the expenditure quintile explains only a minimal fraction of the variance in inflation between households within each year. Most of the observed heterogeneity could be due, among other factors, to differences in the composition of the basket associated with household characteristics not included in this model (such as the type of dwelling, the age of the main breadwinner or the place of residence); however, as it is not part of the specification used, this work cannot confirm its specific contribution to residual variance.

The model with year-to-year interactions reveals the underlying pattern to this effect: the sign of the gap with respect to quintile 1 alternates systematically throughout the period (see Figure 3). The most extreme coefficients are concentrated in 2023, with a difference of +1.40 p.p. in quintile 4 and +2.02 p.p. in quintile 5 with respect to quintile 1 (in both

cases), coinciding with the peak already identified in both the Q1-Q5 gap and the dispersion index. $p < 0,001$

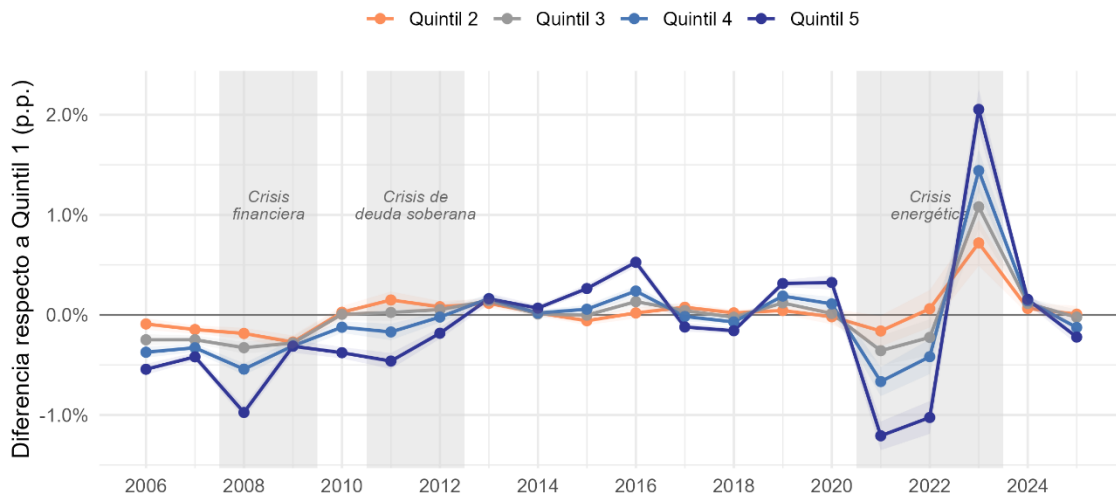


Figure 3. Gap with respect to Quintile 1, per year.

The low *Within R*² of the aggregate model has a simple explanation: almost all the variance of inflation per household depends on the year, not the quintile. A model that only controls for each year, without including the quintile at all, already explains 77.71% of the total variance (*Adj. R*² = 0.7771), practically the same as the 77.73% of the aggregate model, which does include the quintile with a fixed coefficient for the entire period, a very small difference, consistent with the fact that the *Within R*² of the latter is so low. Allowing the quintile effect to vary from year to year, on the other hand, does provide a relevant improvement in the quintile's ability to explain the residual variation: the *Within R*² of the model with interactions (3.22%) is almost thirty times higher than that of the aggregate model (0.11%), although this improvement is barely translated into the *overall Adj. R*² (78.42% vs. 77.73%), since the latter is dominated by the effect of the year common to all quintiles.

This confirms that the quintile effect is not a constant shift over time: by not fixing the years, the model with interactions captures that its effect changes sign from one year to another, consistent with the alternation already documented in the Q1-Q5 gap; while the aggregate model averages all those years in a single number, and much of that information is lost in the average.

In any case, it is important to distinguish between the statistical significance of the quintile effect (confirmed for quintiles 4 and 5, which have significantly lower inflation than 1 in the aggregate model) and the proportion of the residual variation that the quintile explains, which is small. This does not detract from the validity of the conclusions of this

work: it indicates that there are other factors, such as some household characteristics not included in this model, that could explain a large part of the residual variation not captured by the quintile, without contradicting the existence of a significant quintile effect, although of modest magnitude, especially marked at the upper end of the expenditure distribution.

4.3. Mechanism

Knowing that inflationary inequality exists, and that the composition of the consumption basket is, by construction, the only channel through which it can be manifested with the data used (see methodology), this chapter examines in which categories of expenditure this differential composition is concentrated and how much each one contributes to the observed gap. To this end, the difference in the weight that this category occupies in the consumption basket of quintile 1 versus quintile 5 as shown historically in Figure 4 is calculated, as well as the contribution of this category to the total inflation gap between the two quintiles.

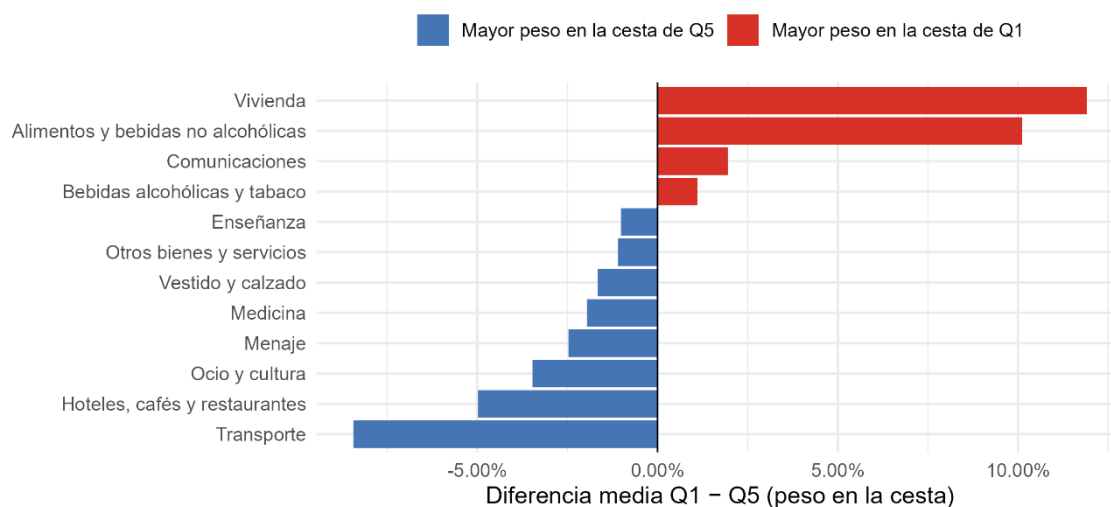


Figure 4. Historical average difference in weight of each category between Q1 and Q5

The most marked differences are concentrated in Housing (+0.120, i.e. Q1 allocates proportionally more expenditure to this category than Q5) and Food and non-alcoholic beverages (+0.101, in the same direction), while Transport presents the opposite sign and the third highest magnitude (-0.084, Q5 proportionally allocates more expenditure to this category than Q1). The rest of the categories show considerably smaller differences, including Hotels, cafes and restaurants (-0.050) and Leisure and culture (-0.035). This pattern is consistent with Engel's law set out in the theoretical framework: households

with the lowest expenditure allocate a greater proportion of their budget to basic necessities (Housing, Food), while those with the highest expenditure allocate proportionally more to categories such as Transport, Leisure or Catering.

Figure 5 transfers this differential composition to its contribution to the inflation gap, multiplying each basket difference by the real inflation of the category in each year. Housing is the category that contributes the most, on average, to penalizing Q1 compared to Q5 (+0.330 p.p.), followed by Food and non-alcoholic beverages (+0.250 p.p.); Transport, on the other hand, contributes in the opposite direction and with the third highest magnitude (-0.168 p.p.), penalizing Q5 on average more, consistent with its greater weight in the basket of the latter quintile. Hotels, cafes and restaurants have a similar magnitude to that of Transport (-0.162 p.p.), also in the opposite direction to Q1; the rest of the categories contribute considerably lower magnitudes, all below 0.06 p.p.

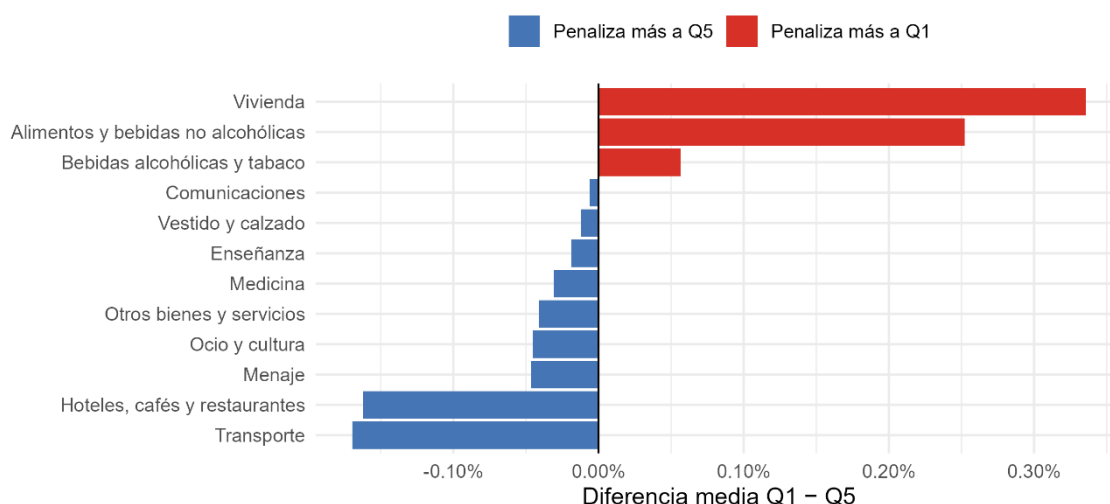


Figure 5. Historical average impact of each category on the Q1-Q5 gap

Taken together, these results show that the inflationary inequality documented in section 4.1 responds to a clear pattern of expenditure composition: the categories where Q1 concentrates proportionally more budget (Housing, Food) are the same as those that penalize Q1 the most compared to Q5, while the categories where Q5 spends proportionally more (Transport, Hotels and Catering) operate in the opposite direction. Since, as explained in the methodology, the composition of the basket is the only channel through which this inequality can manifest itself with the data used, these results are consistent with an entirely compositional mechanism, without prejudice to the heterogeneity of prices within categories that this approach does not allow to capture.

4.4. Aggregation bias

Now it is worth asking what would have happened if we had used a level of product aggregation more common in the literature, such as the single-digit COICOP, with which most official inflation indicators are published. This chapter compares the Q1-Q5 gap obtained with the COICOP-4 detail used in the rest of this work with the one that would be obtained with the single-digit COICOP.

Figure 6 compares the Q1-Q5 gap calculated with the COICOP-4 detail used in the rest of this work compared to the one that would be obtained with the single-digit COICOP. The gap at the detailed level exceeds in magnitude (absolute value) the one calculated at the aggregate level in 10 of the 20 years of the series, a less systematic pattern than that predicted by Jaravel (2021), with an average percentage increase over the 20 years of 155%, which is possibly strongly distorted by the years in which the aggregate gap is close to zero, where any difference translates into a disproportionate percentage increase.

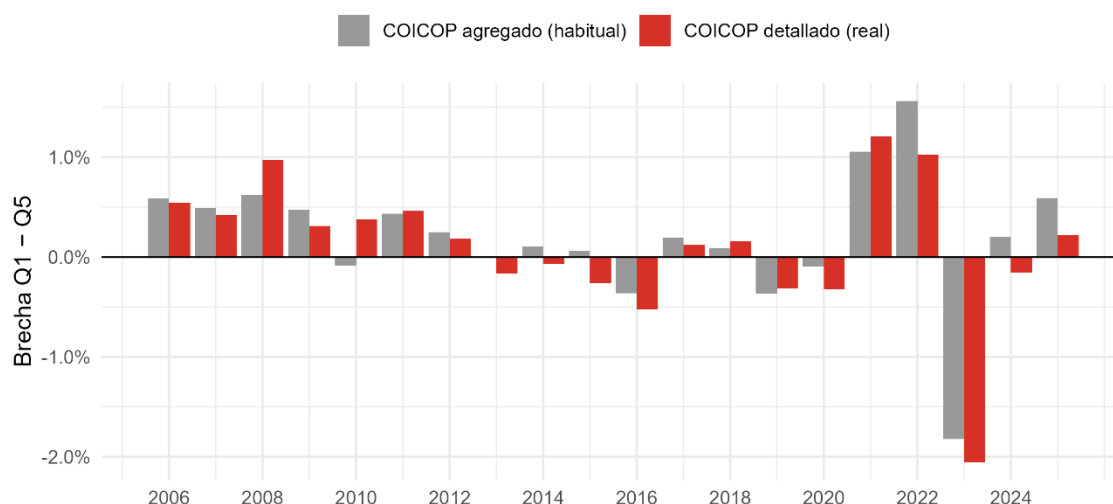


Figure 6. Aggregation bias in the Q1-Q5 gap

This result, weaker than expected according to theory, is consistent with the actual coverage of data at the maximum detail level: only 49.7% of household expenditure has inflation calculated directly at the COICOP-4 level; the rest resort to more aggregate levels because, as explained in the methodology, more than half of the COICOP-4 combinations distinguished by the EPF (224 out of 430) lack their own price category in the CPI. Given that section 4.3 shows that households with lower expenditure allocate proportionally more budget to Housing, and that it is precisely within this division that a significant part of the codes without correspondence in the CPI (including the total

imputed rent) are concentrated, it would be expected that the detailed-aggregate gap estimated here would change appreciably if inflation were available at level 4 for all expenditure on Housing, although this paper does not have the necessary data to quantify this difference. This coverage limitation artificially dilutes the observable difference between the two levels, suggesting that the actual aggregation bias could be greater than that estimated here if level 4 inflation were available for all inflation that is combined with expenditure.

Figure 7 replicates the comparison for the dispersion index among all households, the weighted standard deviation, rather than being limited to the extreme quintiles. Here the pattern is even more consistent: in the 20 years of the series, without exception, the dispersion calculated at the detailed level exceeds that calculated at the aggregate level, with an average increase of 54.3%. This uniformity, compared to the more irregular pattern observed in the Q1-Q5 gap, suggests that aggregation bias systematically affects the dispersion of the distribution as a whole, beyond the two extreme quintiles.

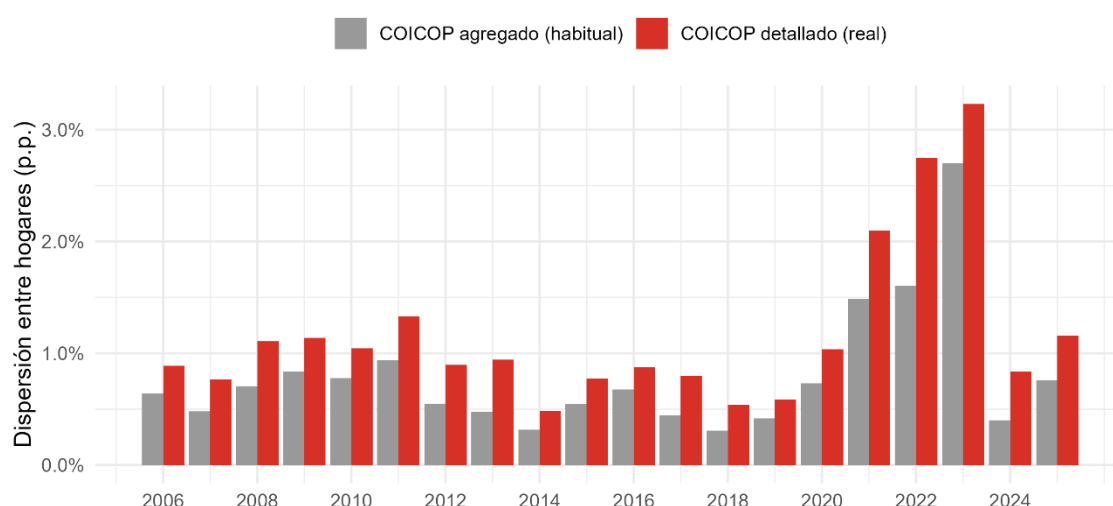


Figure 7. Aggregation bias in dispersion between households

An additional pattern deserves to be highlighted: the difference between the detailed level and the aggregate level is not evenly distributed throughout the series, but is concentrated precisely in the years with the greatest overall dispersion. The three years with the largest absolute difference between the two levels: 2022 (+1.14 p.p.), 2021 (+0.62 p.p.) and 2023 (+0.53 p.p.), coincide exactly with the three years with the greatest dispersion in the entire series, both at the detailed and aggregate levels. In other words, the aggregation bias is not only systematic as a whole, but, at least in this case, it is exacerbated precisely in the episodes of greatest inflationary stress, when the

heterogeneity of prices between products within the same category is greatest. This would suggest that the underestimation of inflationary inequality derived from the use of aggregate indicators is most serious precisely when it matters most: during inflationary crises.

4.5. Policy simulation: counterfactual without VAT reduction on basic foodstuffs

To assess the distributional effect of the VAT reduction on basic foodstuffs (RDL 20/2022), a counterfactual scenario was constructed that reconstructs the inflation that households would have experienced in the absence of this measure. To do this, the inflation of exempt products was recalculated and the resulting Q1-Q5 gap was compared with the gap actually observed. Figure 8 shows both series for the period 2021-2025: the real gap and the gap under the counterfactual scenario without VAT reduction.

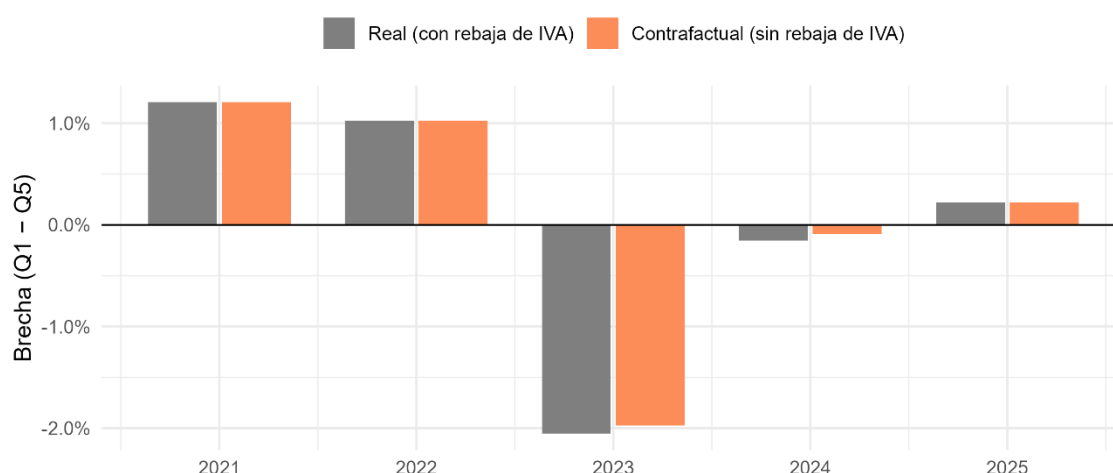


Figure 8. Q1-Q5 gap: real vs. Counterfactual scenario

In 2023, the VAT reduction did not reduce the gap between Q1 and Q5, as might be expected, but widened it slightly: although both quintiles benefited from the policy, Q1 did so to a greater extent (+0.176 p.p. of avoided inflation, from -0.645% to -0.821%, compared to +0.1 p.p. for Q5, from 1.33% to 1.23%), precisely because it allocates a greater proportion of its budget to exempt products. This greater relief for Q1 did not translate into a rapprochement between the two quintiles, but rather the opposite: since it was Q1 that suffered significantly less inflation that year, benefiting the latter, which was already in a better relative position, increased the distance between the two, rather

than reducing it. Thus, the gap in absolute value would have been slightly smaller without the policy (1.97 p.p.) than with it (2.05 p.p.), an adjustment of just 0.08 points.

In 2024, on the other hand, both quintiles would have experienced a somewhat lower rise in inflation in absolute terms than in 2023: +0.15 p.p. for Q1 (from 2.85% to 3.00%) and +0.08 p.p. for Q5 (from 3.01% to 3.09%), after weighting the current tax rate in each part of the year. As in 2023, Q1 benefited somewhat more than Q5 from the policy, and Q5 started from having suffered higher inflation, the gap between the two in absolute value widened slightly: from 0.09 p.p. without the reduction to 0.16 p.p. with it, an adjustment of just 0.07 points.

Translating these effects into euros, the average household in Quintil 1 saved, thanks to the VAT reduction, around €2.39 per month in 2023 (€28.68 per year) and €2.21 per month in 2024 (€26.5 per year), compared to €4.11 and €3.29 per month, respectively, for the average household in Quintil 5 (€49.32 and €39.48 per year). These annual figures are of the same order as those estimated by Almunia, Martínez and Martínez (2023), €35 for lower-spending households and €85 for high-expenditure households, although somewhat lower, especially at the upper end; and significantly lower than those of López Laborda, Marín González and Onrubia (2023), which place the six-month savings of the bottom quintile at around €22-23, compared to the €14.34 that would be obtained here for a comparable period (calculated as half of the annual savings of 2023, given the solely annual resolution of the expenditure data used in this work, which prevents the first half of the year from being specifically isolated). These differences are attributable in the first study, at least in part, to the fact that it compares quintiles instead of deciles; and in the second to the fact that they do not apply a transfer rate at prices of 90% (Almunia, Martínez & Martínez, 2023), in addition to the fact that they use income to build quintiles and not expenditure.

However, these modest figures in absolute terms confirm the conclusion, shared by the two studies, that the impact of this measure is limited compared to the magnitude of the inflationary shock to which it was intended to respond. In absolute euros, the savings were greater for Q5 than for Q1, a mostly mechanical result given that Q5 spends more in absolute terms on these products. But, as a proportion of its own monthly expenditure, the relief was greater for Q1: 0.18% in 2023 and 0.15% in 2024, compared to 0.10% and 0.08%, respectively, for Q5, consistent with the progressive nature of the measure documented in these same studies.

5. Conclusions

This work was based on a question with three dimensions: how much inflationary inequality varies between Spanish households over time, where this variation comes from, and what effect the fiscal policies applied have on it. The results obtained help to answer each of them.

Regarding the magnitude and evolution of inflationary inequality, this paper documents that inflation does not affect the different levels of expenditure equally. Restricting the analysis to the period 2006-2021 to compare it with Basso, Dimakou and Pidkuyko (2022), the average annual gap is 0.194 p.p., very close to the 0.2 p.p. that these authors report for Spain, and its cumulative (3.10 p.p.) is of the same order as the 2.75 p.p. that they document. Restricting the analysis to 2006-2015, the subperiod most comparable to the one used by Jaravel (2019) for the United States, the levels of inflation by quintile obtained in this work (2.40% for the lowest quintile, 2.12% for the highest) are also close to theirs (2.20% and 1.86%, respectively), which constitutes an additional approximate consistency with the magnitude of the documented effect, although the methodological differences between studies indicated in the theoretical framework advise against interpreting this closeness as a strict validation.

However, this gap does not follow a sustained or constant trend over time: its sign alternates between years, with a particularly marked peak coinciding with the energy crisis of 2021-2022, followed by a strong reversal of sign in 2023, with no stable pattern between quintiles observed throughout the series. This alternation is consistent with the idea that inflationary inequality manifests itself as a succession of specific episodes, linked to the price shock at any given time, rather than as a structural and constant bias towards one end of the distribution.

As for the mechanism that causes this inequality, this paper shows that, given that the inflation of each expenditure category comes from the national CPI (common to all households regardless of their quintile), the composition of the consumption basket is the only channel through which inflationary inequality can manifest itself with the data used. Housing (+0.335 p.p.) and Food (+0.252 p.p.), the categories where households with the lowest expenditure proportionally concentrate more budget (with basket differences of +0.119 and +0.101, respectively), are also the ones that penalize this group the most compared to those with higher spending; Transport, where the opposite occurs (basket difference of -0.084 and contribution of -0.169 p.p.), operates in the opposite direction.

In addition, this work provides evidence that the level of aggregation of the data used may condition the magnitude of the inequality detected. When comparing the results obtained with the COICOP-4 detail versus the COICOP-1, the aggregation bias is systematically documented in the dispersion index among all households: the dispersion calculated at the detailed level exceeds the aggregate in the 20 years of the series, without exception, with an average increase of 54.3%. However, this effect is weaker in the Q1-Q5 gap, where the detailed level only exceeds the aggregate in magnitude in 10 of the 20 years analysed. This discrepancy between the two metrics suggests that the aggregation bias, documented by Jaravel (2021) mainly for the gap between extremes, is manifested in this work in a more robust way when the dispersion of the distribution as a whole is considered, and is consistent, in the case of the Q1-Q5 gap, with the incomplete coverage of data at the maximum detail level available for Spain.

Finally, in terms of policies capable of mitigating this inequality, the simulation of the VAT reduction on basic foodstuffs (RDL 20/2022) shows a real distributional effect, albeit modest in absolute terms. In 2023, the policy reduced inflation in quintile 1 by 0.18 percentage points compared to the scenario without a reduction, compared to 0.10 p.p. for quintile 5; in 2024, with both quintiles registering positive inflation, the relative relief remained in the same direction, although of considerably smaller magnitude (4.9% compared to 2.6%). In terms of the Q1-Q5 gap, however, this relief did not translate into a reduction in inequality between the two quintiles: by benefiting proportionally more the quintile that already suffered less inflation each year, the policy slightly widened the distance between the two (from 0.08 p.p. in 2023 and 0.07 p.p. in 2024), an effect contrary to what could be done, although of almost insignificant magnitude compared to the total disparity between quintiles in each year. In euros, monthly savings ranged between €2 and €4 per household depending on the quintile and year, figures of the same order of magnitude as those estimated by other studies on this same measure, although somewhat lower and not directly comparable given that they use different groups, periods and translation assumptions, as detailed in chapter 4.5. These results suggest that this type of measure has a limited scope compared to the total magnitude of inflationary inequality documented in this work.

5.1. Limitations

This work has several limitations that should be noted. Firstly, although the COICOP-4 level of detail used substantially reduces the aggregation bias compared to the most common levels in the literature, it is still a product aggregation: within the same level 4

category, products with different prices and inflation dynamics (for example, different brands or types of establishment) can coexist, a phenomenon of the same nature as the documented aggregation bias, only operating on an even finer scale. This limitation could only be overcome with price data at the individual product level, such as the scanner data used by Kaplan and Schulhofer-Wohl (2017), who show that most of the price heterogeneity between households comes precisely from this source, and not from differences in the composition of the basket between categories.

Secondly, the coverage of the CPI at the maximum detail level is not complete: as detailed in the methodology, 224 of the 430 COICOP-4 combinations distinguished by the EPF (52%) lack their own price category in the CPI, a stable limitation throughout the series and not concentrated in any specific time frame; for these categories, this work resorts to a cascade aggregation process, imputing inflation from the next higher level. This solution makes it possible to maintain complete coverage of the consumption basket, but introduces, within the "detailed" level used in this work, an aggregation fraction that is not completely eliminated. This could explain why this bias, documented in section 4.4, is weaker in the Q1-Q5 gap than the literature predicts. It would be desirable, for future extensions of this work, to have complete CPI series at the COICOP-4 level for all years and categories, which would allow estimating the real aggregation bias without this coverage limitation.

Thirdly, the policy simulation is based on comparative statics: it reconstructs the inflation that each household would have experienced without the VAT reduction, keeping the consumption basket observed in reality fixed, without incorporating how households themselves would have reacted to this hypothetical absence of the measure. Specifically, the possible substitution of products documented in the literature on the effects of fiscal policies is not contemplated: a household could have reduced its consumption of basic foodstuffs or replaced some of them with cheaper alternatives if their price had risen more, without the reduction, than it actually did, and vice versa.

Finally, the EPF collects household expenditure on an annual basis, while the CPI is published on a monthly basis. This asymmetry forces this paper to average the monthly inflation of each category throughout the year to obtain a single annual figure, thus losing all the price variation that occurs within the same year. This limitation would be especially relevant for the fiscal policy analysis developed in chapter 4.5: the VAT reduction on basic foodstuffs was not in force with the same tax rate throughout 2024 (with a gradual de-escalation in the last quarter), and although this work incorporates an approximation to this change by weighting the effective rate by the number of months in each bracket,

it cannot accurately capture how inflation evolved month by month, nor contrast the reaction of household spending in real time to the rate change. In the same way that adding products with different price dynamics under the same COICOP category hides relevant heterogeneity, adding months with different inflation levels under the same annual figure could hide equally relevant temporal heterogeneity, especially in episodes of high price volatility such as those analyzed in this work.

5.2. Future lines

This work suggests several lines of extension. First, the role of other sociodemographic characteristics of the household as additional factors of inflationary heterogeneity could be explored, along the lines of what has already been explored by Basso, Dimakou and Pidkuyko (2022) for Spain: the size of the household, the age or educational level of the main breadwinner, the type of home tenure (ownership versus renting) or the type of employment contract could interact with the expenditure quintile to explain part of the residual heterogeneity that this work attributes to unobserved household characteristics. An advantage of this work is that the data cleansing process, as mentioned in the methodology, has already included these variables, so that the study of these variables would not entail reprocessing the data from the INE's PFS.

Secondly, it would be relevant to extend the analysis to the territorial dimension, given that the Bank of Spain itself has documented regional heterogeneity in the composition of the basket and in the inflation borne by autonomous community. The PFS includes the variable of autonomous community of residence of each household, so it would be possible to replicate the methodology of this work at the regional level, exploring whether the Q1-Q5 gap documented at the national level hides different patterns within certain autonomous communities, particularly relevant given that some fiscal and housing competencies are decentralized.

Third, a natural extension of this work, conditional on the availability of the appropriate data source, would be to use *scanner data* at the barcode level in Spain, following the approach of Kaplan and Schulhofer-Wohl (2017) for the United States. This type of data, usually collected from the cash registers of large distribution chains, would make it possible to observe the exact price and brand of each product purchased by each household, and not just its aggregated COICOP category. With this information, it would be possible to construct an inflation per household that also captures the heterogeneity of prices within each level 4 category. In Spain, this line of research would depend on the availability of agreements for access to scanner data from large distribution chains,

which are currently non-existent in the national academic sphere, unlike the already consolidated availability of this type of data in the United States.

Finally, a natural extension of the policy simulation would be to incorporate the effects indicated in the constraints such as product substitution, as well as to apply the same methodology to other fiscal policies in response to inflation, such as the reduction in the price of electricity already evaluated by Basso, Dimakou and Pidkuyko (2022), which would make it possible to systematically compare the distributive efficiency of different public policy instruments in the face of the same inflationary shock.

Overall, this paper provides three specific contributions to the literature on inflationary inequality in Spain. First, it documents that the inflation gap between high- and low-spending households does not follow a sustained trend, but rather manifests itself as a succession of episodes linked to specific crises (particularly markedly, the energy crisis of 2021-2022; and, in particular, the reversal of sign observed in 2023), with magnitudes that show an approximate consistency with those reported by the international and Spanish reference literature. Second, it documents, applying one of the finest levels of product disaggregation used to date in the Spanish literature on the subject, that this inequality is consistent with a fundamentally compositional mechanism, concentrated in Housing and Food, and that the level of aggregation of the data used conditions the magnitude of the inequality detected, in a particularly robust way in the dispersion of the distribution as a whole. Third, it estimates the distributional effect of the VAT reduction on basic foodstuffs for the entire period of validity of the measure (2023-2024), including its gradual de-escalation, a longer time horizon than that covered by previous evaluations of this same policy (Almunia, Martínez, & Martínez, 2023; López Laborda, Marín González, & Onrubia, 2023), integrating the simulation into a broader analysis of inflationary inequality built with a level of product detail unusual in the available Spanish literature, and suggesting that its design was progressive in relative terms, although of modest magnitude compared to the scale of the inflationary shock to which it responded, lower than that estimated by these previous studies, and without translating, in practice, into a reduction in the Q1-Q5 gap.

Artificial Intelligence Usage Statement

In the preparation of this Final Degree Project, generative artificial intelligence has been used as a one-off support tool, mainly for the refinement of the writing, the debugging of errors in the code and the generation of repetitive code (such as the construction of graphics). In no case has it been used to generate the substantive content, results or conclusions of the work.

ChatGPT (free version, with the GPT-5.6 Luna model), developed by OpenAI, has been used.

The use of this tool was limited to three types of support tasks: (1) suggestions for improvement in the clarity and fluidity of the writing of certain paragraphs, without altering the substantive content or conclusions; (2) help to identify and correct errors (*bugs*) in the R code used for microdata cleaning and statistical analysis; and (3) generation of repetitive code fragments, mainly for the elaboration of graphs, from specifications provided by the author. In all cases, the result proposed by the tool was reviewed, verified and, when necessary, corrected by the author before its incorporation into the work.

To guarantee the academic integrity and originality of the work, the author has personally reviewed and validated all the content generated with the support of this tool, contrasting any code suggestions with the results obtained and verifying that the editorial corrections did not alter the meaning or rigor of the original statements. All methodological decisions, the design of the analysis, the interpretation of the results and the conclusions of the work are of her own elaboration.

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Appendices

Table A1. Comparison of average annual expenditure per household (own elaboration) with official figures from the INE

Year	Average expenditure (own elaboration)	Average expenditure (INE)	Difference (%)
2006	30.232,91	30.278,78	-0,15
2007	31.661,90	31.701,36	-0,12
2008	31.731,48	31.772,62	-0,13
2009	30.207,01	30.245,09	-0,13
2010	29.453,28	29.492,15	-0,13
2011	29.147,81	29.187,00	-0,13
2012	28.159,90	28.197,00	-0,13
2013	27.114,24	27.150,75	-0,13
2014	27.054,16	27.089,83	-0,13
2015	27.436,15	27.473,04	-0,13
2016	28.201,26	28.199,88	0,00
2017	29.188,19	29.188,19	0,00
2018	29.871,28	29.871,28	0,00
2019	30.242,76	30.242,76	0,00
2020	26.998,36	26.995,76	0,00
2021	29.243,61	29.243,61	0,00
2022	31.567,71	31.567,71	0,00
2023	32.616,66	32.616,66	0,00
2024	34.044,03	34.044,00	0,00
2025	35.100,71	35.101,00	0,00

Source: INE figures in the table "Total expenditure, average expenditure and distribution of household expenditure". Figures for 2024 and 2025 from the INE press releases corresponding to the final results of the EPF for each year (National Institute of Statistics, 2025, 2026-b).